UVARAM MOHANARAM

Software Engineer & AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Software Engineering undergraduate at General Sir John Kotelawala Defence University specializing in Artificial Intelligence, Machine Learning, MLOps, and Full-Stack Development. Experienced in architecting end-to-end intelligent systems—from fine-tuning multilingual transformer encoders (`int8` ONNX) and PyTorch neural networks to deploying low-latency FastAPI inference pipelines and cloud-native Next.js/React applications. Proven track record leading technical teams in national AI/ML buildathons alongside university honours as a KDU Chess Colours recipient, international FIDE-rated chess player, and university athlete.

EDUCATION

General Sir John Kotelawala Defence University (KDU), Sri Lanka — **2025 – Present**
BSc (Hons) in Software Engineering — Faculty of Computing (Intake 42 • Reg. No. D/BSE/25/0012)

Hindu College, Colombo — **Completed 2024**
Secondary Education — Served as School Prefect and Athletics Captain

TECHNICAL SKILLS

AI / ML & Data Science	PyTorch (`torch.nn`, MLP, Custom Training), Hugging Face Transformers (`multilingual-e5-small`), ONNX Runtime (Dynamic Int8 Quantization), scikit-learn (TF-IDF, Linear SVM, `ColumnTransformer`), Pandas, NumPy, Pinecone Vector DB, RAG
Programming Languages	Python, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3
Frameworks & Backend	FastAPI, Next.js (App Router, Server Actions, API Routes), React, Vite, Tailwind CSS, Node.js, RESTful APIs, OpenAPI 3.0 & JSON Schema Validation, WebSockets, Progressive Web Apps (PWA)
Cloud, DB & MLOps	Supabase (PostgreSQL, Row-Level Security, Realtime), SQLite, Docker, Microsoft Azure (VMs, Monitor), Caddy (HTTPS Reverse Proxy), Vercel, Git, GitHub Actions (CI/CD), Git LFS, Jira, Google Maps APIs, Stripe API

AI / ML & SOFTWARE ENGINEERING PROJECTS

TensorForge 2.0 (Phase 2 MVP) — Multilingual Support Ticket Classification & Routing — **Oct 2026**
Role: Team Leader (Team Fade) | **Stack:** Python, Hugging Face, ONNX Runtime, scikit-learn, FastAPI, SQLite, Docker, Azure, Caddy | GitHub • Live Service • Demo • Report

- Engineered a CPU-only, offline-capable multilingual ticket classification and routing service across English, Sinhala, Tamil, Singlish, and Tanglish (11 primary categories, secondary category, urgency, and human-review thresholding across 4,800 tickets).
- Fused a word-and-character TF-IDF Linear SVM with a fine-tuned `intfloat/multilingual-e5-small` encoder exported to dynamically quantized int8 ONNX, elevating development-validation macro-F1 from 0.6558 to 0.8022 (+14.63 pts) and urgency F1 to 0.9080.
- Built an asynchronous FastAPI + SQLite batch job worker handling single predictions (p95 35ms), synchronous batches (1–100 tickets), and large evaluation jobs (up to 5,000 tickets), deployed via Docker and Caddy HTTPS on Azure VM with 256/256 contract checks passing in GitHub Actions CI.

AI-Powered Dynamic Public Transport Fleet Optimizer (Route 100) Oct 2026

Role: ML & Backend Engineer | **Stack:** PyTorch, scikit-learn, FastAPI, Pandas, NumPy, Joblib | Technical Report & Evidence

- Designed and trained a 4-layer PyTorch neural network (`DemandPredictorMLP`: 23 → 128 → 64 → 32 → 1) with `BatchNorm1d`, `Dropout (0.2)`, ReLU activations, Adam optimizer, and Huber Loss (Smooth L1) to predict continuous passenger demand across 8 Galle Road bus stops.
- Built a leak-proof `ColumnTransformer` pipeline (`MinMaxScaler` on 11 calendar/temporal features + `OneHotEncoder` on 3 categorical features) with strict chronological 80/20 date splitting across 800,000 records (158,622 holdout test rows; MAE 225.8, RMSE 519.3).
- Integrated the PyTorch model into an asynchronous FastAPI service with vectorized `torch.no_grad()` batch inference to dynamically calculate bus allocations and dispatch headways across rush-hour, off-peak, holiday, and special-event overrides (7/7 verification scenarios passed).

DropMe — Decentralized Corridor Micro-Transit Optimization System Aug – Sep 2026

Role: Team Leader & Lead Developer | **Stack:** React, Vite, Tailwind CSS (PWA), Supabase (PostgreSQL, RLS, Realtime), Google Maps APIs | Pitch Deck • Proposal

- Architected a corridor-based commuter micro-transit Progressive Web App (PWA) with ACID-compliant Supabase seat locking and sub-second WebSocket state synchronization to convert empty commuter seats into scheduled campus and urban shuttles.
- Engineered a deterministic funnel-topology route-matching engine using Google Maps Distance Matrix, Places Autocomplete, and Directions APIs, guaranteeing zero drop-off detour and a strict waypoint pickup detour budget ($\Delta T \leq 5$ min).
- Designed a distance-proportional billing model and a tiered `.ac.lk` institutional KYC and live-selfie verification pipeline with burn-after-approval privacy lifecycle, tripling average vehicle occupancy (1.2 → 3.5+ riders) and cutting per-capita CO_2 emissions by 55–65%.

UN Tiles — Intelligent E-Commerce & Inventory Management Platform Jul – Sep 2026

Role: Full-Stack Developer | **Stack:** Next.js, TypeScript, Tailwind CSS, Supabase (PostgreSQL), Pinecone, Stripe, Vercel | GitHub • Implementation Report • System Design

- Developed a full-stack e-commerce and real-time inventory management platform for architectural tiles using Next.js (Server Actions & API Routes), TypeScript, Tailwind CSS, Supabase PostgreSQL, Pinecone vector search (visual tile matching), and Stripe checkout.
- Built the `/api/seed` migration route, refactored user profiles into an authenticated "Studio Account Portal," and implemented a smart tile wastage calculator (10–15% margin estimation) and automated low-stock threshold alerts.
- Configured GitHub repository workflows, environment variables, and Vercel CI/CD production deployments across agile sprints tracked in Jira.

HONOURS, SPORTS & LEADERSHIP

- **KDU Chess Colours Awardee (2025) & International FIDE Rated Player:** Awarded KDU University Colours for Chess (2025); active international FIDE-rated tournament player (FIDE ID: 9923489).
- **KDU University Athletics & Sports Leadership:** Active member of the General Sir John Kotelawala Defence University Athletics contingent; former School Prefect, Athletics Captain, and Karate competitor at Hindu College, Colombo.
- **AI & Engineering Competitions (Team Leader — Team Fade):** Led university teams in IEEE CS KDU TensorForge 2.0 AI/MLOps Buildathon (2026), CodeSplash '26, IEEE SIGHT SDG Sprints 2026, and GDG Sri Lanka Build with AI Buildathon (2026).
- **Languages:** Tamil (Native) • English (Advanced Professional Proficiency) • Sinhala (Advanced Professional Proficiency)